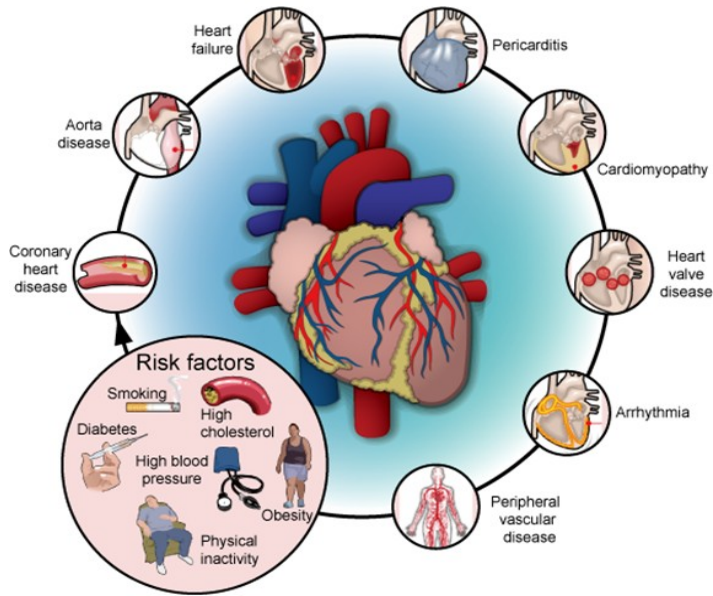
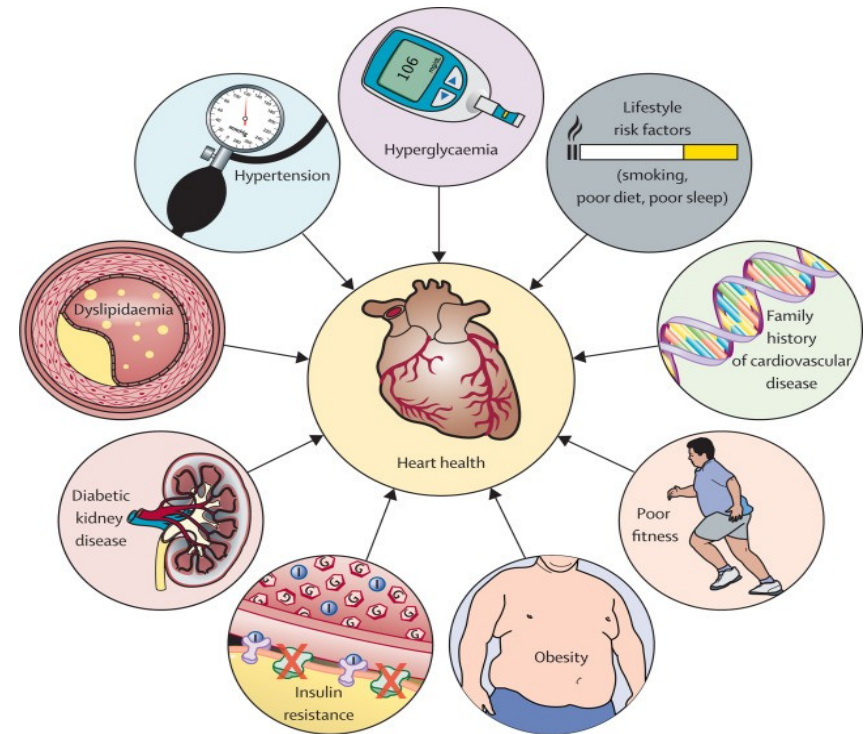


HEART DISEASE



Cardiac emergency

Dr. Hanaa Elsayed



Learning Outcome

- By the end of this lecture, students should be able to describe:
 - **Coronary Artery Disease (CAD):**
 - Define CAD
 - Enumerate risk factors
 - List clinical pictures
 - **Hypertension:**
 - Define hypertension
 - Enumerate risk factors
 - List clinical pictures
 - **Heart failure:**
 - Define heart failure
 - Enumerate risk factors
 - List clinical pictures
 - Management plan for patients with cardiac emergencies includes CAD, hypertension, and heart failure.
 - Describe dental aspect for patients with cardiac emergencies

Introduction

- Heart is vital to life.
 - **Function:** is pumping oxygenated blood all over the body and vital organs (brain).
- **Cardiovascular disease (CVD)** is a major killer and causes **morbidity and disability**.
- **Cardiovascular (CV) emergencies** are **rare in dental practice**.
- **Dentists must be able to:**
 - **Identify medical emergencies** which may be life-threatening
 - **Needs appropriate skills**, training and **available equipment** to treat them quickly and effectively.

- **It includes:**
 - **Coronary artery disease (CAD) includes angina pectoris and myocardial infarction (MI)**
 - **Hypertension, heart failure and stroke**
 - **Cardiac arrhythmias or dysrhythmias**
 - **Infective endocarditis**
- **About half of all medical emergencies occur during dental procedures as injections of local anesthesia, during extractions or pulp extirpations (eradication).**

- **Once cardiac arrest is recognized:**



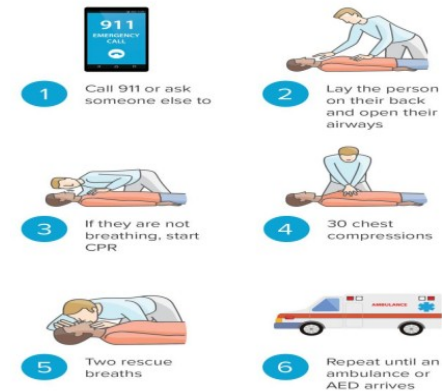
- Dental team must immediately call emergency medical services (EMS)
- Chest compressions, clearing airways, and restoring breathing (CAB)

- **Dental chair** is a good for **chest compression** because it provides stable support.



- All **dental team should** receive basic life support training.
- Keep **emergency drugs** and **equipment** available, and **train staff** how to use them.

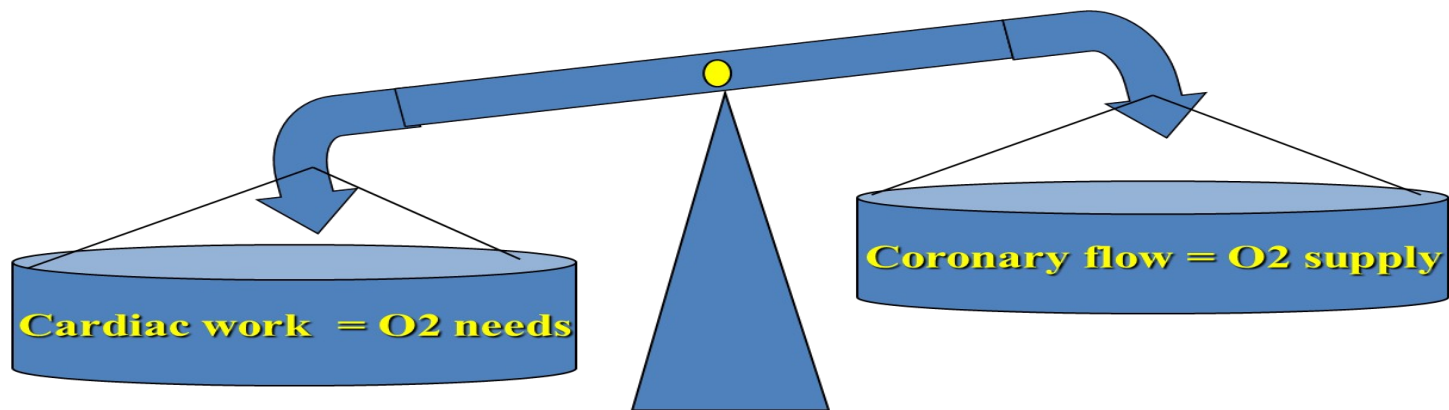
CPR: Step by Step



<i>Drug</i>	<i>Indication</i>	<i>Initial Adult Dose</i>
1. Oxygen	almost any medical emergency	100%: inhalation
2. Epinephrine	anaphylaxis	0.1 mg i.v., or 0.3–0.5 mg i.m.
	asthma unresponsive to albuterol/salbutamol	0.1 mg i.v., or 0.3–0.5 mg i.m.
	cardiac arrest	1 mg i.v.
3. Nitroglycerin	pain of angina	0.3–0.4 mg sublingual
4. Antihistamine (diphenhydramine or chlorpheniramine)	allergic reactions	25–50 mg i.v., i.m.
		10–20 mg i.v., i.m.
5. Albuterol/salbutamol	asthmatic bronchospasm	2 sprays: inhalation
6. Aspirin	myocardial infarction	160–325 mg

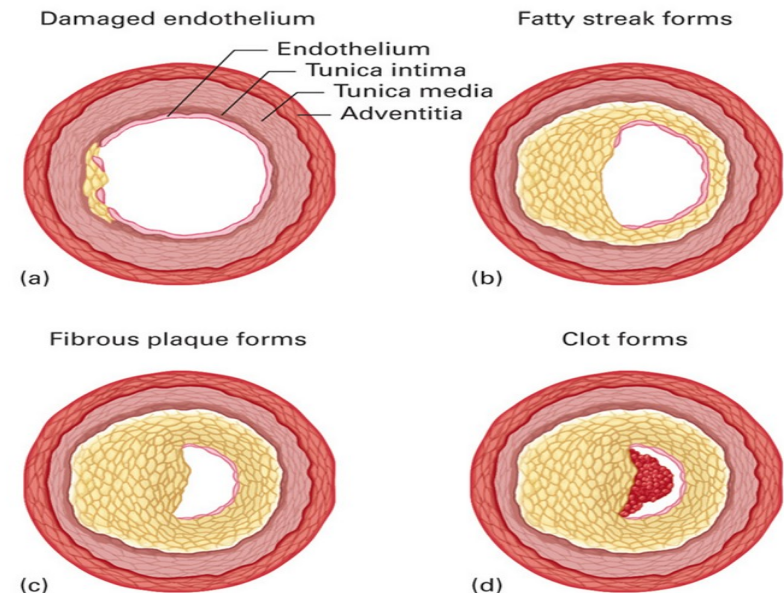
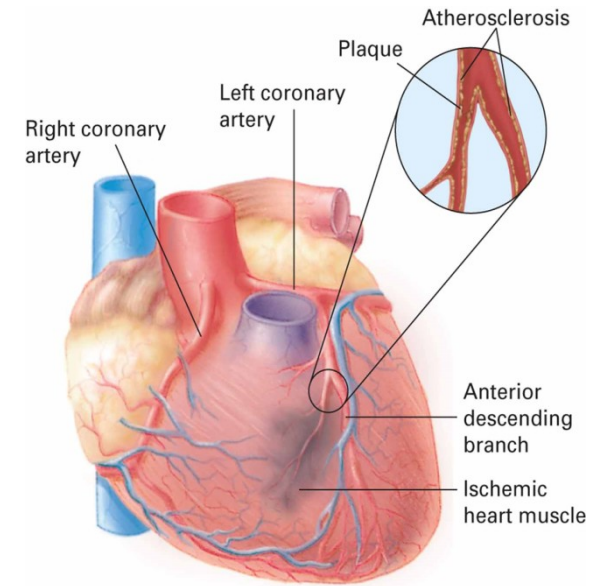
<i>Drug</i>	<i>Indication</i>	<i>Initial Adult Dose</i>
1. Glucagon	hypoglycemia in unconscious patient	1 mg i.m.
2. Atropine	clinically significant bradycardia	0.5 mg i.v. or i.m.
3. Ephedrine	clinically significant hypotension	5 mg i.v., or 10–25 mg i.m.
4. Hydrocortisone	adrenal insufficiency	100 mg i.v. or i.m.
	recurrent anaphylaxis	100 mg i.v. or i.m.
5. Morphine or nitrous oxide	angina-like pain unresponsive to nitroglycerin	titrate 2 mg i.v., 5 mg i.m.
	angina-like pain unresponsive to nitroglycerin	~35%, inhalation
6. Naloxone	reversal of opioid overdose	0.1 mg i.v.
7. Lorazepam or Midazolam	status epilepticus	4 mg i.m. or i.v.
	status epilepticus	5 mg i.m. or i.v.
8. Flumazenil	benzodiazepine overdose	0.1 mg i.v.

- **Definition:** Cardiac ischemia is due to **imbalance between** oxygen supply and demand in the myocardium.
 - **Factors that decrease** myocardial oxygen supply:
 - Decreased luminal diameter: **atherosclerosis, vasospasm, anemia.**
 - **Factors that increase** myocardial oxygen demand:
 - Hyperthyroidism, myocardial hypertrophy, aortic stenosis



Epidemiology

- CAD is the most common cause of cardiovascular morbidity and mortality.
- **Male: female** ratio = 2:1
- **Age:** 50-60 (men) and 60-70 (women).
- **Pathogenesis:**
 - **Atherosclerosis and thrombosis** are the most important pathogenic mechanisms.



Risk factors

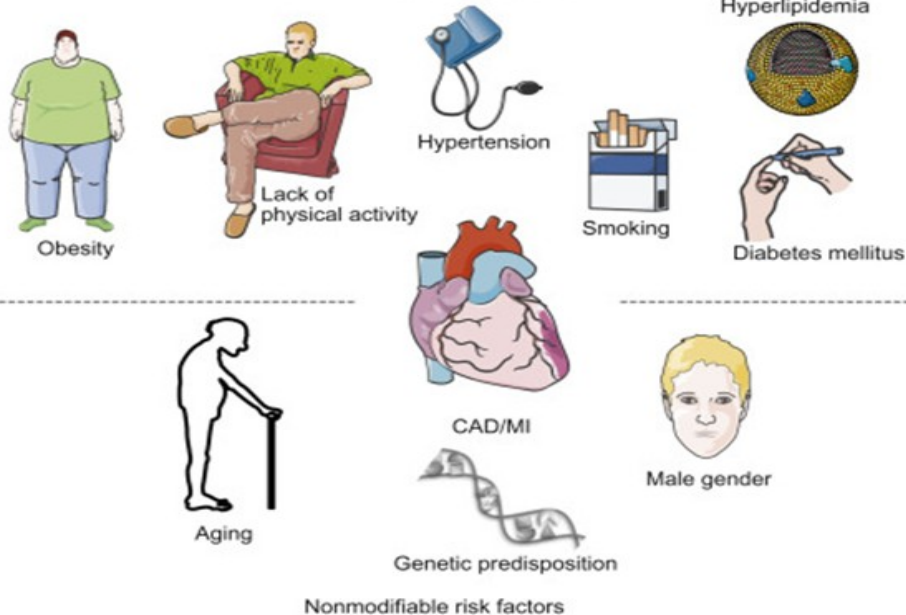
- **Non-modifiable:**

- **Age.**

- **Male.**

- **Family history of**

Modifiable risk factors



- **Modifiable risk factor:**

- Hyperlipidemia.
- Hypertension (HTN).
- Diabetes mellitus (DM).
- Cigarette smoking.
- Obesity, stress, sedentary life.
- Heavy alcohol intake



Signs and Symptoms

- Typical **retrosternal chest pain**.
- **Tightness or discomfort** in character.
- **Radiated to left** ($\pm$ right) arm, neck, jaw, teeth or back.
- **Associated with** dyspnea, diaphoresis (sweating), nausea, anxiety.
- **Duration:** lasting 10-<15 min.
- **Precipitated by “3 Es”:**
 - Exertion, emotion, eating.
- **Relieved by** rest and nitrates.



Diagnosis

- History and examination.
- Laboratory diagnosis:

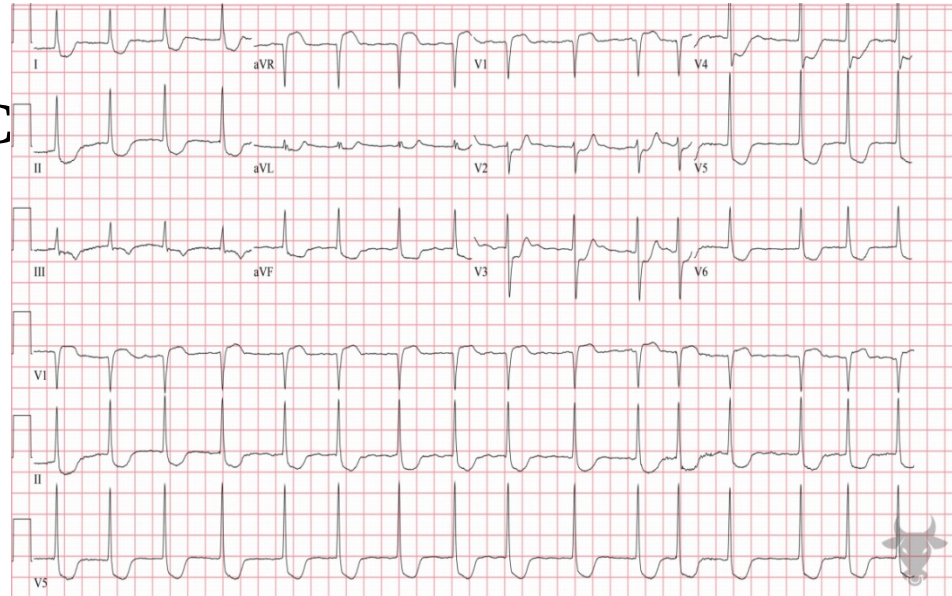
- Complete blood count (CBC)
- Blood glucose level
- Lipid profile.

- Radiological:

- Electrocardiograph (ECG):

- Depressed ST, inverted-T changes.

- Plain X-ray chest (CXR).
- Echocardiography.

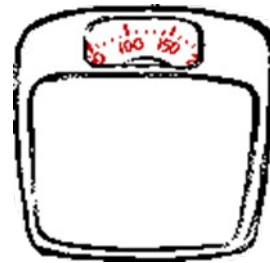


Treatment of Angina Pectoris

- **General Measures (non pharmacological):**

- **Life style modification:**

- Stop smoking
 - Gradual exercise
 - Weight reduction



- **Avoid:**

- Exertion
 - Emotions
 - Eating heavy fatty meals
 - Exposure to cold.

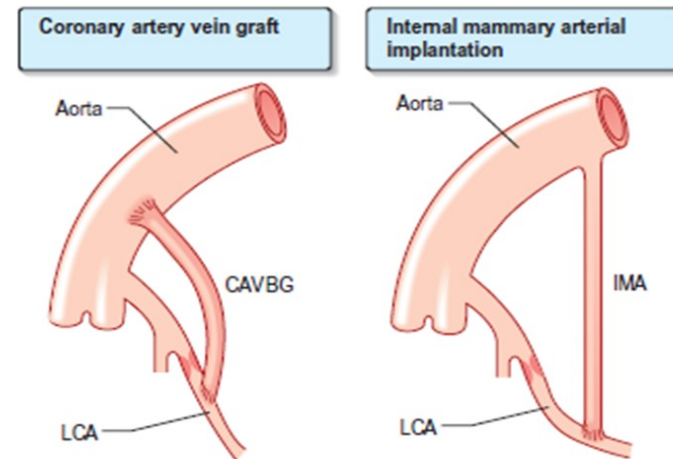
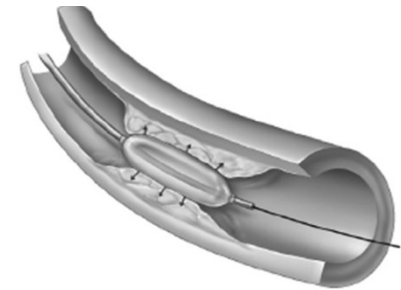
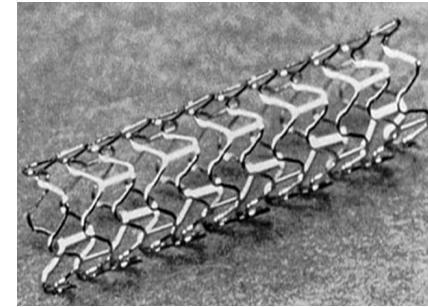


- **Pharmacological treatment**

- **Antiplatelet** as Aspirin, clopidogrel.
- **Lipid-lowering therapy** (statin).
- **Glyceryl trinitrate** (GTN) used sublingually, oral, skin patch or spray.
 - **Adverse effects:**
 - Throbbing headache, flushing, and dizziness, and interfere with **sildenafil**.
- **Beta-blockers**, nitrates and calcium-channel blockers or combination therapy.
- **Revascularization if not responding to treatment**

Surgical treatment

- **Percutaneous coronary intervention (PCI):** is the process of dilating a coronary artery stenosis **using an inflatable balloon** introduced via the femoral, radial or brachial artery.
- **Coronary artery bypass graft (CABG)** **autologous veins or arteries** are anastomosed to the ascending aorta and to the native coronary arteries **distal to the area of stenosis.**

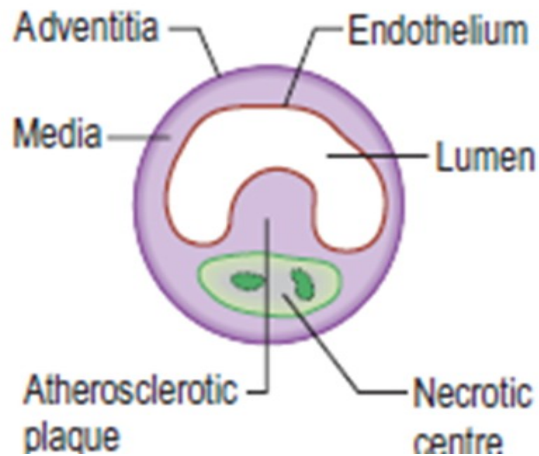


Acute Coronary syndrome (ACS)

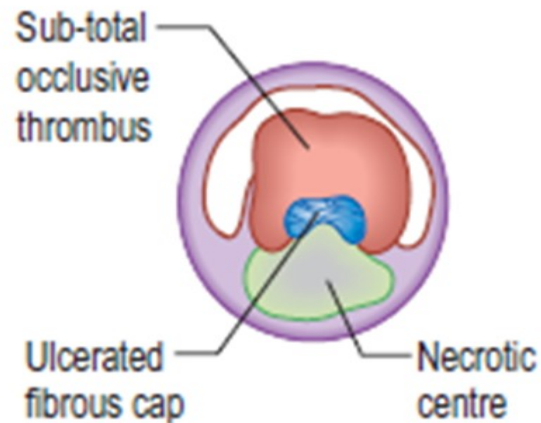
- ☠- Emergency case //
- ACS includes:
 - Unstable angina (UA).
 - Non S-T segment elevated myocardial infarction (NSTEMI) and
 - S-T segment elevated myocardial infarction (STEMI).
- MI is defined by:
 - Elevated cardiac markers
 - ECG changes (elevated ST segment with inverted T wave or pathological Q waves)

Relationship between the state of coronary artery vessel wall and clinical syndrome

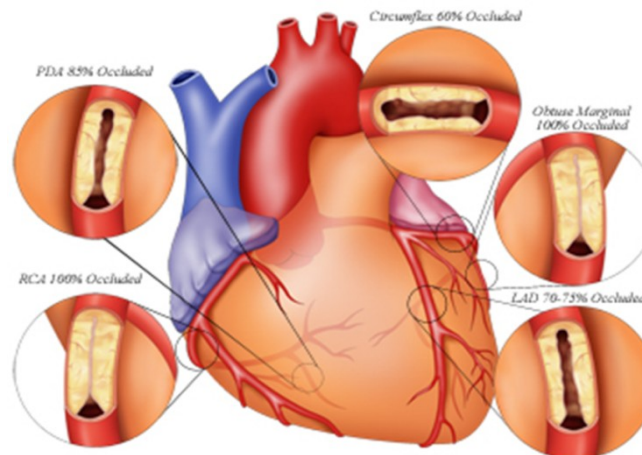
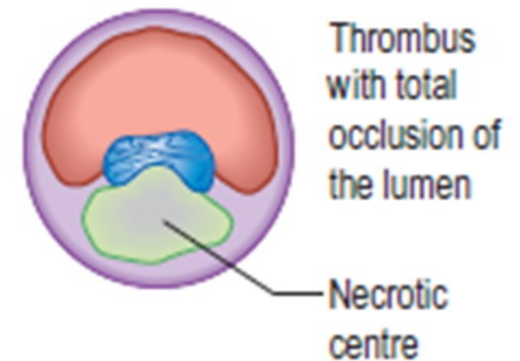
(a) Stable angina pectoris



(b) Unstable angina pectoris



(c) Myocardial infarction (STEM I)

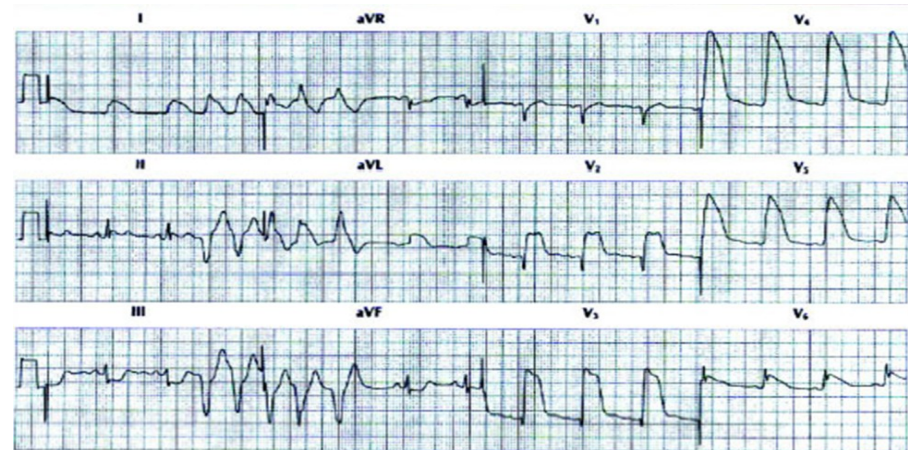


Clinical Pictures

- Central chest pain similar to angina **but** more severe, longer-lasting, **not relieved by rest or sublingual (SL) nitrates.**
- Characterized by sense of tightness, crushing, compression or constriction of the chest.
- It sometimes **radiates to the left arm or jaw.**
- **Associated with** dyspnea, nausea and vomiting.

Diagnosis

- History and physical examination.
- ECG.
- Plain X ray chest (CXR).
- **Laboratory investigations.**
 - Cardiac biomarkers for myocardial damage:
 - Troponin I & T.
 - CK-MB.
 - Myoglobin (repeat 8 h later).
 - CBC, INR/PTT.
 - Creatinine, urea.
 - Blood Glucose levels.
 - Serum lipids.



ECG showing S-T segment elevated myocardial infarction

Management of ACS

- Bed rest, cardiac monitoring.
- Coronary vasodilatation:
 - Glycerial trinitrate.
- Analgesia: Morphine IV.
- Oxygen.
- Antiplatelet: Aspirin, Clopidogril.
- ↓ Myocardial energy consumption: β blocker.
- Plaque stabilization/ventricular remodeling:
 - Statins, ACE inhibitors.

- **Plus for ST-elevation myocardial infarction (STEMI):**
 - **Immediate reperfusion therapy:**
 - **Thrombolytic therapy:**
 - Streptokinase
 - Alteplase Or
 - PCI.
 - **Most common cause of death after MI is fatal arrhythmia**

Dental aspects of angina

- There is association between periodontal disease and cardiovascular disease.
- Angina pain radiating to mandible mimic pain of dental origin.
- Stress, anxiety precipitate angina pain (enhanced sympathetic activity and adrenaline/epinephrine release) therefore, patients should receive dental care in short, minimally stressful appointments as late morning.
 - Early morning or late afternoon appointments avoided in IHD patients

- **Pre-operative** GTN and oral anxiolytics, nitrous oxide or sedation (e.g. temazepam) are advised.
- **Local anesthesia** with long-acting as Bupivacaine **minimize the release of endogenous catecholamine.**
 - Use **local anesthesia** containing **low** concentration (1:100,000 or less) of epinephrine **with caution.**

- **General anesthesia** should be avoided for dental care in cardiac patients (recent angina and unstable angina) at least 3 months **unless essential or in emergency** and must be given in hospital with cardiac consultation.
- **Any dental surgery within 6 months of MI** should be avoided due to high risk of complication **as MI, arrhythmias.**
 - Elective surgery should be deferred during this duration.

- **Antiplatelet or anticoagulant** can cause **excessive bleeding** during dental procedure, **use Local hemostatic techniques.**
 - **If patient on anticoagulants,** International Normalized Ratio (INR) should be measured **within 24 hours before the procedure.**
 - **A minor dental surgical procedure** can be done **when INR of 2–4 with use of local hemostatic measures to control bleeding.**
 - **If INR more than 4,** no surgical intervention performed **without cardiac consultation.**

- Aspirin contraindicated in severe hypertension or failure due to salt and fluid retention.
- Indomethacin may interfere antihypertensive agents.
- Erythromycin and azoles should be avoided in patient taking statins due to increased muscle damage.
- Intramuscular injection of drugs should be avoided because absorption may be too slow and increases the risk of local bleeding.

- **Drugs may cause oral adverse effects:**
 - **Angiotensin converting enzyme inhibitors (ACEIs) cause ulcers and taste changes.**
 - **Beta-blockers causing lichenoid lesions** (bilateral white papules forming reticular or annular plaque-like causing a burning sensation, bleeding, and pain).
 - **Calcium-channel blockers (nifedipin) causing gingival swelling.**

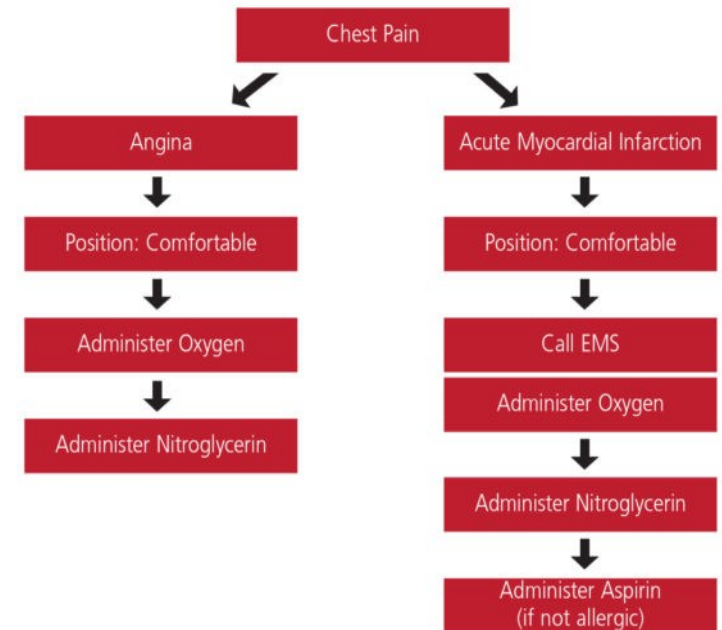
Oral Mucosal lichenoid lesions



- **If patient develops chest pain during dental procedure:**
 - **Dental treatment should be discontinued.**
 - **Keep patient in sitting upright.**
 - **Vital signs should be monitored.**
 - **Short-acting nitrate (GTN sublingually) with oxygen.**
 - **Pain should be relieved in 2–3 min. GTN can be given every 5 minutes until the pain improve.**
 - **If not improved:**
 - **Call emergency department.**

- **MI is a possible cause if:**
 - Chest pain is not relieved within 3 min., lasts more than 15–20 min.
 - If pain persists after 3 doses of nitroglycerin/5 min.
 - Associated with nausea, vomiting, syncope or hypertension.
 - **Treatment:**
 - Continue oxygen, and chew 300 mg of aspirin.
 - Sent them to hospital.

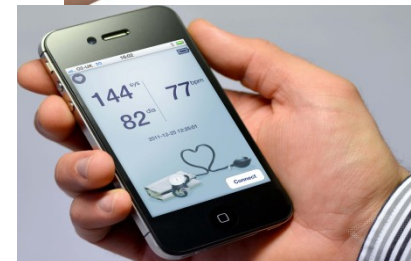
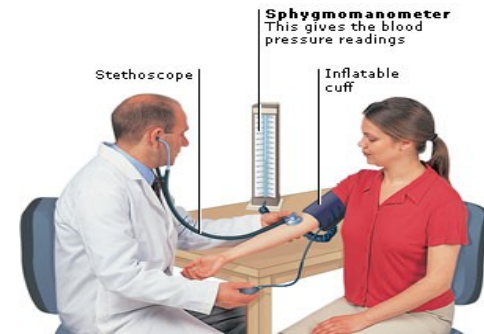
TABLE 3. Steps to Follow When a Patient Experiences Chest Pain



EMS: emergency medical services.

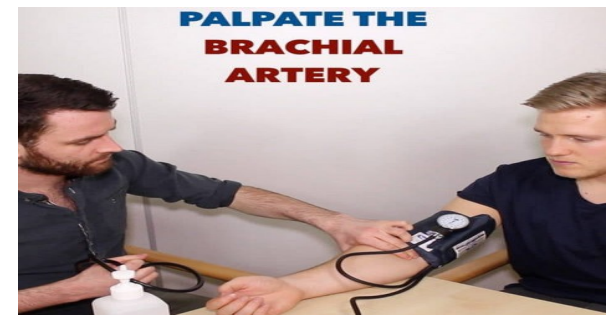
Systemic hypertension (HTN)

- **Definition:** persistent elevation of blood pressure (BP) $\geq 140/90$ mmHg.
- **Prevalence** of HTN about **30–45%** of the general population. Less than half of adults (42%) with hypertension are diagnosed and treated.
- Approximately 1 in 5 adults (21%) with hypertension have it under control.
- **HTN** is a **major cause** of premature death worldwide. It can **leading to** MI, heart failure (HF), chronic kidney disease (CKD), and Stroke.



Manual technique for recording blood pressure

1. Ask patient to avoid alcohol and smoking for 30 min before measurement.
2. Allow patient to sit at rest for as long as possible.
3. Place sphygmomanometer cuff on right upper arm with about 3 cm of skin visible at antecubital fossa; bladder should encircle at least two thirds of arm.
4. Palpate right brachial pulse.
5. After 5 min, inflate cuff slowly to about 200–250 mmHg, or until pulse is no longer hearing.
6. Deflate cuff slowly while listening with stethoscope over brachial artery below cuff.
7. Record systolic pressure as pressure when first tapping sounds appear (Korotkoff sounds).
8. Deflate cuff further until tapping sounds become muffled (diastolic pressure) and then disappear.
9. Record BP as systolic/diastolic pressures.



Stages of hypertension

- {European Society of Hypertension (ESH)}
- Optimal BP <120/80
- Normal BP:
 - Less than 120/80 mmHg.
- Elevated BP (high normal) or Pre-hypertension:
 - BP 130–139/85–89 mmHg
- Stage 1 hypertension:
 - BP $\geq$ 140-159/90-99 mmHg
- Stage 2 hypertension:
 - BP $\geq$ 160-179/100-109 mmHg.
- Severe hypertension:
 - Systolic BP $\geq$ 180 mmHg and/or diastolic BP $\geq$ 110 mmHg.

Diagnosis	Systolic (top)	Diastolic (bottom)
Normal	<120	<80
Prehypertension	120-139	80-89
Stage 1 Hypertension	140-159	90-99
Stage 2 Hypertension	>160	>100
Hypertensive Emergency	>180	>120

- **American College of Cardiology (ACC) and the American Heart Association (AHA) has now separated hypertension into 2 categories:**
 - **Normal BP <120/80**
 - **Elevated (SBP 120–129 mmHg and DBP < 80 mmHg)**
 - **Stage 1 {>130/80 mmHg} (SBP 130–139 mmHg or DBP 80–89 mmHg).**
 - **Stage 2 hypertension BP $\geq$ 140/90 mm Hg.**

Etiology and risk factors

- **Primary or essential:** > 90-95%
- **Secondary:** < 5-10 %
- **Essential hypertension (90%-95%):** unknown cause.

- **Genetic Factors:**

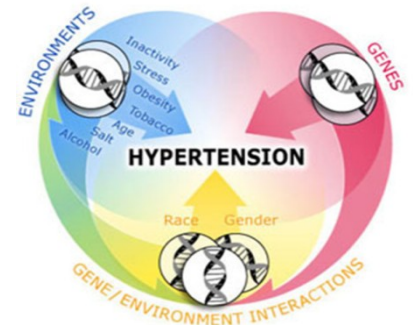
- » HTN more common in some families (positive family history).
- » Ethnic groups like African Americans.

- **Environmental factors** include:

- » Obesity, lack of exercise and excess salt, fat intake.
- » Stress, tobacco, alcohol.

- **Sympathetic over-activity:**

- » **40% of hypertensive patients** have raised catecholamine (adrenaline/epinephrine and noradrenaline/norepinephrine)



Secondary hypertension

- **Renal:** over 80 % of the cases of secondary HTN:
 - Renal artery stenosis.
 - Pyelonephritis.
 - Glomerulonephritis(GN).
 - Polycystic kidney disease (PKD).
- **Endocrinal:**
 - Pheochromocytoma, Acromegaly, Cushing's syndrome.
- **Drugs:**
 - Oral contraceptive pills, corticosteroids, NSAIDs.
- **Pregnancy**-induced hypertension.
- **Vascular** as vasculitis.

Clinical pictures and complications

- Hypertension usually has no warning signs or symptoms.

- HTN affect many tissues:

- **Brain:** headache, visual disorders and stroke.
- **Heart:** CAD, MI, heart failure (HF).
- **Eyes:** hypertensive retinopathy or optic neuropathy and blindness.
- **Kidneys:** proteinuria and progressive renal insufficiency.

Organ	Problem
	Blood vessel
	Atherosclerosis Aneurysms Aortic dissections
	Chronic kidney disease
	Haematuria Uraemia Proteinaemia
	Cardiac failure
	Pulmonary oedema Myocardial infarction Left ventricular hypertrophy
	Stroke / TIA
	Haemorrhage / infarction Seizures Vascular dementia
	Retinal damage
	Haemorrhages Exudates A-V nipping Papilloedema

Diagnosis

- History and examination.
- Ambulatory blood pressure monitoring (ABPM) and Home BP monitoring (HBPM) are needed to confirm diagnosis of HTN.
- **Routine investigation** of the hypertensive patient should include:
 - Fasting blood for total cholesterol (target <200 mg/dL) and high-density lipoprotein (target ≥ 60 mg/dL), LDL (target <100 mg/dL) and triglycerides (target <150 – 199 mg/dL).
 - Fasting blood glucose level.
 - Serum urea, creatinine and electrolytes.
 - Urine.
 - ECG.
 - Chest x-ray.



Management

- **Non-pharmacological treatment:**
 - **Weight reduction:** physical activity (30 minutes /day, 5 days/week).
 - **Decrease alcohol consumption:** ≤ 21 units/week for **men** and ≤ 14 for **women**.
 - **Eating healthy diet:**
 - Increased fruit, vegetable and fish oil consumption.
 - Low-fat, saturated fat diet and low salt diet.
 - **Smoking cessation.**
 - **Reduce anxiety, stress and emotion.**

Pharmacological treatment

- **Antihypertensive therapy (monotherapy or combination)** is given with minimum dose and minimal adverse effects.
- **Several drugs are available to treat HTN includes:**
 - Angiotensin converting enzyme (ACE) inhibitors or Angiotensin receptor antagonists (ARBs).
 - Beta-blockers as atenolol.
 - Calcium-channel blockers.
 - Thiazide- diuretics.
 - Other drugs as α -blocker, direct vasodilator, or centrally acting drugs.

Guideline Similarities	2017 ACC/AHA	2023 ESH
Accurate Blood Pressure Measurement	Office-based BP measurements and use of validated, cuffed devices and home/ambulatory BP monitoring are recommended prior to diagnosing hypertension.	
Initial Pharmacotherapy Recommendations	Initial therapeutic choices include ACE inhibitors, angiotensin-receptor blockers, thiazide or thiazide-like diuretics, and calcium channel blockers. Single pill combination therapy is a first-line strategy for many patients.	
Guideline Differences	2017 ACC/AHA	2023 ESH
Hypertension Definition	$\geq 130/80$	$\geq 140/90$
Normal BP Ranges (mmHg)	Normal: $< 120/80$ Elevated: 120-129/<80	Optimal: $< 120/80$ Normal: 120-129/80-84 High-Normal: 130-139/85-89
Hypertensive BP Ranges (mmHg)	Hypertension Stage 1: 130-139/80-89 Hypertension Stage 2: $\geq 140/90$	Hypertension Grade 1: 140-159/90-99 Hypertension Grade 2: 160-179/100-109 Hypertension Grade 3: $\geq 180/110$
BP Targets for Treatment		
18 – 64 years (mmHg)	$< 130/80$	$< 130/80$
65-79 years (mmHg)	$< 130/80$	$< 140/80^*$
≥ 80 years (mmHg)	$< 130/80$	140-150/ <80
Pharmacotherapy	Initial therapy with beta-blockers reserved for specific conditions including ischemic heart disease or heart failure	Beta blockers included as first-line therapy for hypertension.

* Target $< 130/80$ if tolerated

Dental aspects of hypertension

- There are **no recognized orofacial manifestations of HTN.**
- **Initial evaluation of each patient with HTN should include:**
 - History of hypertension, duration and antihypertensive drugs.
 - Severity of disease, and its complications as stroke and facial palsy.
 - Family history of hypertension, cardiovascular disease.

- **Blood pressure (BP) measurement is important at dental visits before dental treatment to reduce complications of:**
 - **Oral surgical procedures.**
 - **Periodontal surgery.**
 - **Dental treatment especially when conscious sedation or general anesthesia is required.**

- According to American Dental Association, outpatient dental care for adult hypertensive patients are as follows:
 - BP ↓ 160/100mmHg: no modification for elective or emergency treatment.
 - If ↑ 160/100 mmHg (repeat measure), avoid elective treatment, and patient should seek consultation with a physician.
 - Initiate dental emergency care if systolic (SBP) 160–180 and/or diastolic (DBP) 100–109mmHg with BP monitoring/10–15 minute during procedure.
 - If SBP > 180 and/or DBP > 109, seek consultation with a physician before procedure if symptomatic.

- Dental procedure should be **postponed** in a patient with **hypertensive crisis** (SBP >180 or DBP > 120 mmHg) and should be referred to emergency care immediately to prevent stroke or end-organ damage.
- Control of pain and anxiety is very important because stressful procedure may increase BP and **causes complications** as cardiac arrest or stroke.
 - Sedation (temazepam) with pre-operative reassurance to control anxiety.
 - Short, minimally stressful appointments (late morning).

- Sudden positional changes during or at the end of prolonged dental procedure **should be avoided** especially old age, those with diabetes mellitus **to prevent postural hypotension.**
- General anesthesia (GA) **potentiate antihypertensive drugs and can induce dangerous hypotension.**



- **Vasoconstrictors** as **epinephrine** combined with **local anesthesia** **cause:**
 - Prolong anesthetic effect.
 - Providing local hemostasis and reduce bleeding.
 - Slow injection can prevent side effect.
 - It is rarely **contraindicated** **unless** SBP is over 200 mmHg and/or the **diastolic** is over 115 mmHg.
- **NSAIDs (indomethacin, ibuprofen and naproxen)** can **reduce or may antagonize** the efficacy of antihypertensive agents and **increase risk of cardiovascular disease** including heart failure.
 - **Paracetamol can be used to avoid** this side effect.

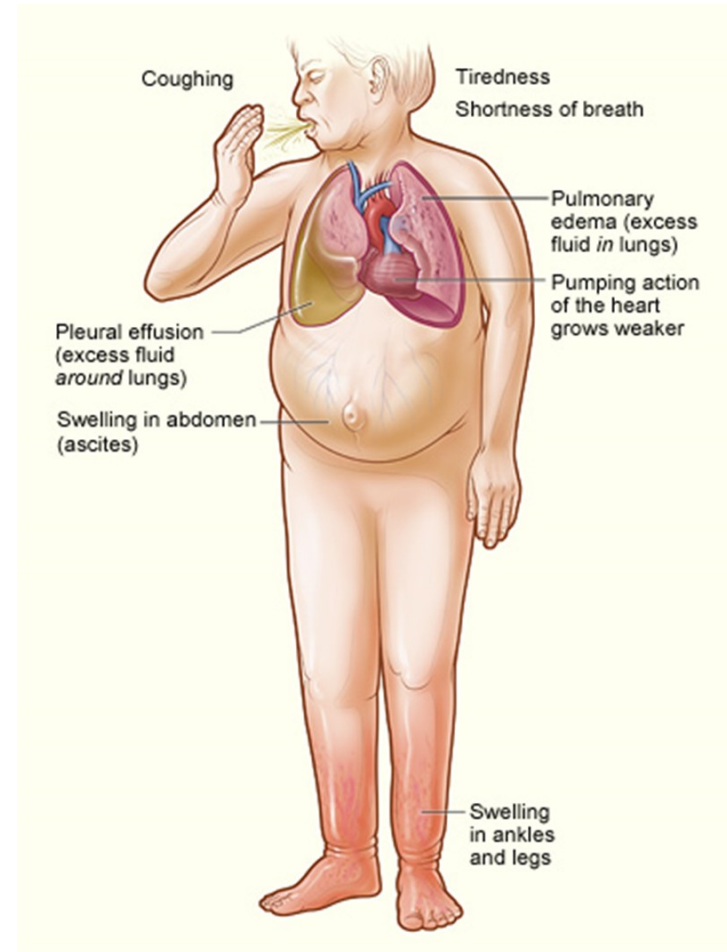
- **Orofacial side effect of antihypertensive medications:**
 - **ACE Inhibitors:** dry cough, oral ulceration, angioedema and alteration or loss of taste sensation.
 - **Beta-Adrenergic Blockers:** xerostomia and oral lichenoid reactions (OLR)
 - **Calcium-channel blocker (nifedipine):** gingival hyperplasia, dry mouth, altered taste.
 - **Diuretics:** Xerostomia, lichenoid reaction
 - **Effect of Xerostomia:** dental caries, halitosis (bad breath), infection, gingivitis and bleeding/ulceration.
 - **Treatment:** drinking more water and using oral lubricants, gums, pastes and sprays

Heart Failure (HF)

- **Clinical syndrome** that can **result from** any **structural or functional cardiac disorder** that **impairs the ability of the heart to function** as a pump to support a physiological circulation.
- **Causes:**
 - **The most common cause of cardiac failure is IHD.**
 - **Myocardial** (cardiomyopathy), valvular (aortic and mitral valve) pericardial, hypertension and arrhythmias.
 - Others as thyroid disease, anemia.

Symptoms & Signs

- **Left heart failure**
 - **Symptoms** are predominantly **fatigue**.
 - Exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea (PND), cough, hemoptysis (coughing of blood) and recurrent chest infection.
 - **Signs:**
 - Cardiomegaly, gallop, functional mitral regurgitation and **crackles or crepitation at the lung bases**.



- **Right heart failure**

- **Symptoms:**

- Dyspepsia, pain in the right hypochondrium (right upper abdominal pain), anorexia and nausea due to systemic venous congestion.



- **Physical signs:**

- **Congested neck vein** (Jugular venous distension).
 - Hepatic enlargement.
 - **Bilateral pitting lower limb edema.**
 - **Ascites**
 - Pleural effusion (transudates on right-sided).
 - Cardiomegaly.



Diagnosis

- **History and examination.**
- **Blood tests:**
 - CBC.
 - Liver biochemistry, urea and electrolytes.
 - Cardiac enzymes:
 - » **BNP [brain natriuretic peptide].**
 - Thyroid function.
- **Chest X-ray.**
- **Electrocardiogram (ECG)**
- **Echocardiography.**

Treatment

- **General measures:**
 - **Rest and stress reduction.**
 - **Salt restriction, stop smoking, weight loss.**
 - **Treat the cause:**
 - **Treatment of hypertension or infective endocarditis.**
 - **Surgical treatment of valvular lesions.**
 - **Treat anemia, thyroid.**

Pharmacological treatment

- ACE inhibitors as enalapril or ARBs as losartan.
- β -blockers.
- $\pm$ Aldosterone antagonists (if severe CHF).
- **Diuretic.**
- $\pm$ Inotrope as digoxin.
- $\pm$ Antiarrhythmic.
- $\pm$ Anticoagulant.

Dental aspect

- Patients with **uncontrolled HF**, **elective dental treatment should be delayed until the condition has been stabilized.**
- The dental chair should be kept in a partially reclining or erect position.
- **Analgesic for pain and anxiety** (precipitate arrhythmias, angina or HF).
- GA is contraindicated in cardiac failure until it is under control.



- **Drugs used with caution as:**
 - **NSAIDs and aspirin:** Increases the risk of renal damage from ACEI.
 - **Itraconazole:** Can aggravate cardiac failure.
- **Drugs may cause oral adverse effects:**
 - **ACEIs:** cause cough, erythema multiforme, angioedema, and burning mouth.
 - **Diuretics:** Orthostatic hypotension.

- **A 73-year-old diabetic patient presented by attacks of retrosternal discomfort radiating to left shoulder precipitated by exertion and relieved by rest after 10 minutes. What is the cause of his complaint?**
- A. Angina.
- B. Pneumothorax.
- C. Costochondritis.
- D. Esophageal spasm.

- **A 65-year-old female patient known to have angina experienced chest pain in dental surgery. What is the proper management in this case?**
- A. Give analgesic and Continue dental procedure.
- B. Give another dose of local anesthesia with epinephrine/adrenaline.
- C. Stop dental procedure and give nitrate [GTN] sublingually.
- D. Use general anesthesia.

- **Which of the following is considered a most common risk factor for systemic hypertension?**
- A. Pheochromocytoma.
- B. Acromegaly.
- C. Cushing's syndrome.
- D. Renal artery stenosis.

- **Which of the following is a good advice regarding the diet of hypertensive patient?**
- A. Low sodium diet.
- B. Increased saturated fat.
- C. Low fruit and vegetable consumption.
- D. Increased simple sugars.



THANK YOU